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| **Status** | Finished |
| **Started** | Wednesday, 28 October 2026, 7:30 PM |
| **Completed** | Wednesday, 28 October 2026, 7:58 PM |
| **Duration** | 27 mins 22 secs |
| **Grade** | **14.00** out of 15.00 (**93.33**%) |

Question **1**

Correct

Mark 1.00 out of 1.00

What happens when the element is found in linear search?

a. The search backtracks to find duplicate elements
b. The search starts over from the beginning
c. The search stops immediately ✓
d. The search continues until the end of the list

Question **2**

Correct

Mark 1.00 out of 1.00

Which method of searching involves sequentially comparing each element until a match is found?

a. Hashing
b. Binary search
c. Linear search ✓
d. Jump search

Question **3**

Correct

Mark 1.00 out of 1.00

_____ search takes a sorted/ordered list and divides it in the middle.

a. Hash
b. Both (1) & (3)
c. Binary ✓
d. Linear

Question **4**

Correct

Mark 1.00 out of 1.00

In linear search, if the target element is not found in the list, what is the result?

- a. The first element is returned
- b. The search is considered unsuccessful ✓
- c. An error is raised
- d. The last element is returned

Question **5**

Correct

Mark 1.00 out of 1.00

In binary search, how is the middle element determined?

- a. By starting from the first element
- b. By comparing each element sequentially
- c. By using a hash function
- d. By dividing the list length by two ✓

Question **6**

Correct

Mark 1.00 out of 1.00

In a linear search, how many comparisons are made in the worst-case scenario to find an element in a list of size n?

- a. n/2
- b. log n
- c. n ✓
- d. 1

Question **7**

Correct

Mark 1.00 out of 1.00

Which of the following best describes the process of a linear search?

- a. Sorting the list before searching
- b. Skipping every second element
- c. Dividing the list in half repeatedly
- d. Checking each element sequentially ✓

Question **8**

Correct

Mark 1.00 out of 1.00

Which of the following scenarios is best suited for applying binary search?

a. When the list contains duplicate elements
b. When the list is sorted ✓
c. When the list is very small
d. When the list is unsorted

Question **9**

Correct

Mark 1.00 out of 1.00

Finding the location of a given item in a collection of items is called

a. Finding
b. Searching ✓
c. Mining
d. Discovering

Question **10**

Incorrect

Mark 0.00 out of 1.00

In binary search, what happens if the middle element does not match the target element?

a. The search stops
b. The list is sorted
c. The search continues from the beginning ✗
d. The search continues in the left or right sublist

Question **11**

Correct

Mark 1.00 out of 1.00

What is searching in the context of computer science?

a. Determining whether an element is present in a list ✓
b. Sorting elements in a list
c. Deleting elements from a list
d. Inserting elements into a list

Question **12**

Correct

Mark 1.00 out of 1.00

Which of the following is not the required condition for a binary search algorithm?

a. The list must be sorted

b. Number values should only be present

Number values should only be present

c. There must be a mechanism to delete and/or insert elements in the list ✓

d. There should be direct access to the middle element in any sublist

Question **13**

Correct

Mark 1.00 out of 1.00

What is the time complexity of binary search in the worst case?

a. O(1)

b. O(n log n)

c. O👎

d. O(log n) ✓

Question **14**

Correct

Mark 1.00 out of 1.00

If a list contains 1000 elements, how many comparisons would a binary search typically make in the worst case?

a. 100

b. 1000

c. 500

d. 10 ✓

Question **15**

Correct

Mark 1.00 out of 1.00

What is the best-case time complexity of linear search?

a. O(log n)

b. O👎

c. O(1) ✓

d. O(n log n)